Module 1 | Introduction to Deep Learning

08 June 2025 20:51

What is Deep Learning ?

Deep learning is a subset of machine learning (ML & DL are subsets of AI) that uses artificial neural network (ANN) to mimic the learning process of the human brain.

ANN

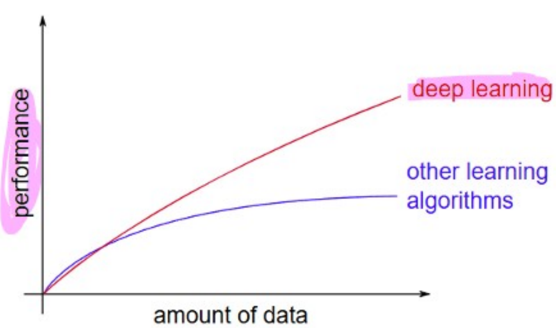
Artificial Neural Network

(man-made neural n/w) → using hardware & software.

software → algorithms using a language PYTHON

NVIDIA
AMD
INTEL
} (GPUs + CPUs)

OS
DBMS
DSA



[CPU vs GPU vs TPU
- wait for module #3]

- ANN is a computational model inspired by the structure and function of the brain.

[there are billions of neurons]

- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

... layers of neurons that
process information and learn patterns from the data.

AI terminologies are not only confusing but are buzzwords?

Disclaimer: No intention to undermine brand's identity.

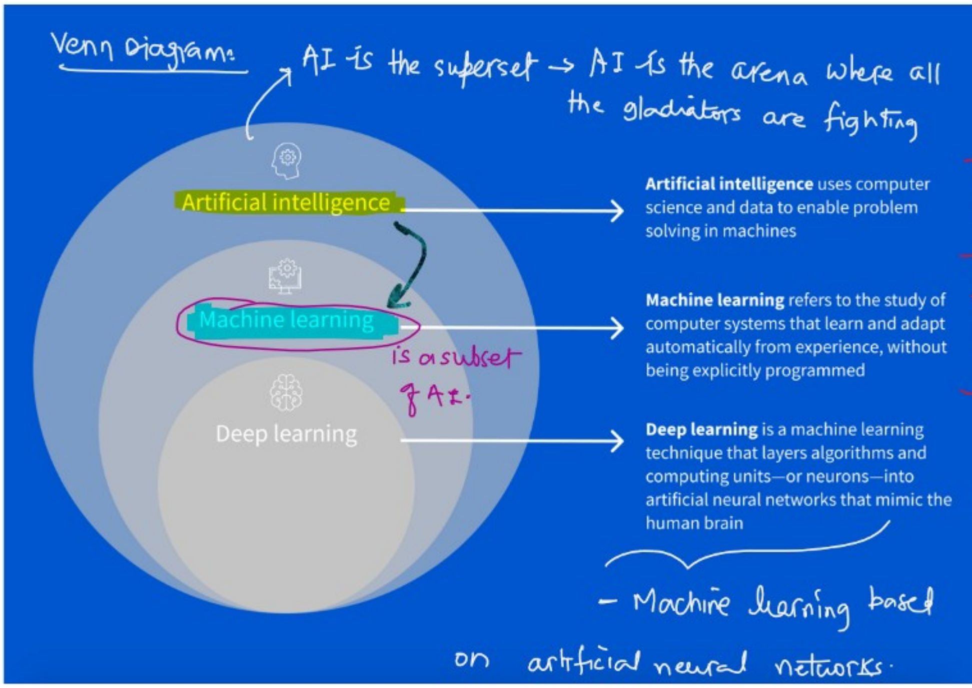
it doesn't mean AI here.

just a marketing gimmick.

↓

AI is needed for sales & marketing.

Difference between AI, ML and DL



AI is the playground to engineer machines empowered by data to mimic congnitive functions (Human Intelligence)

Data is the new oil

Pro-Tip

Software development vs Machine learning

Aspect	Traditional Programming (Chef with strict recipe)	Machine Learning (ML Chef)
Approach	Follows a strict recipe (explicit rules)	Experiments and self-improves (self-adjusting rules)
Example Steps	1. Add 2 cups flour2. Add 1 cup sugar3. Bake at 180°C for 30 min	Bake a cake → Get feedback → Adjust next time
Consistency	Always produces the same result	Result improves over time based on feedback
Ability to Learn from Mistakes	Does not learn from past mistakes	Learns and adjusts based on feedback
Flexibility	Rigid, rule-based	Flexible, adaptive
Outcome over time	Same cake every time	Better cake over time
Nature of Rules	Explicit, hand-coded (IF-ELSE, loops)	Self-adjusting, data-driven
Core Analogy	Chef blindly following a recipe	Chef who learns and adapts based on experience

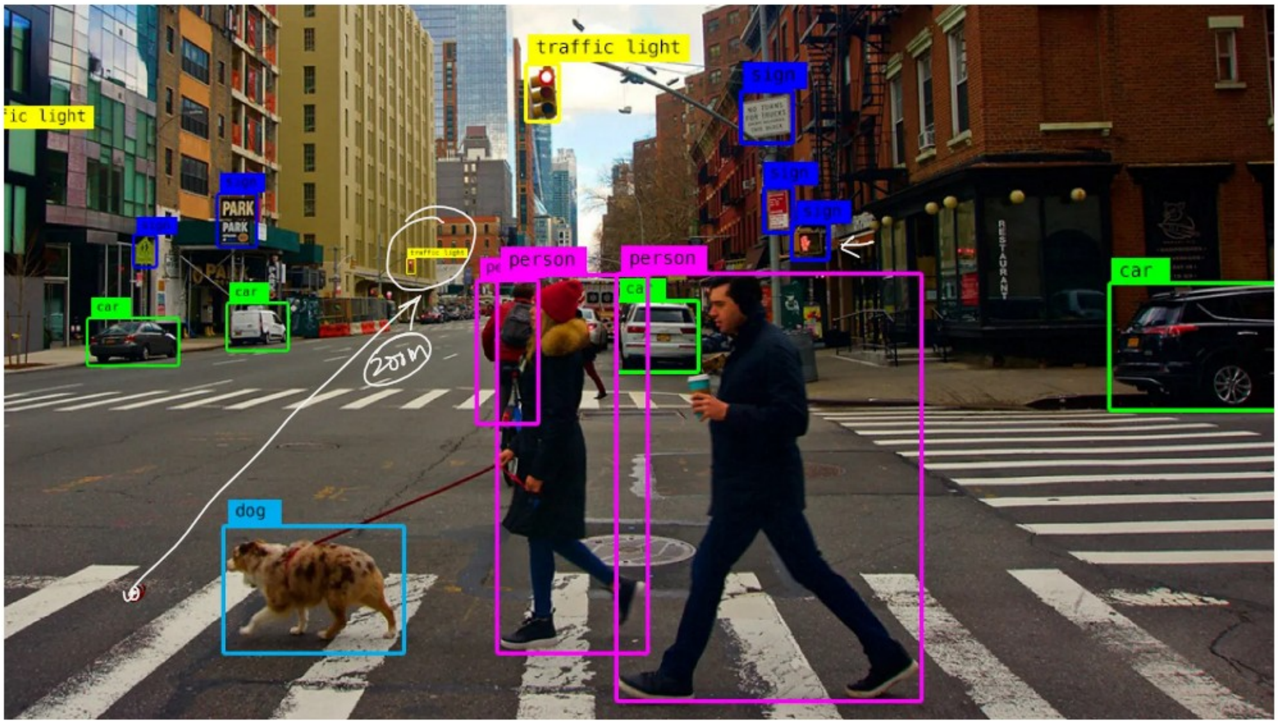
Pro-Tip

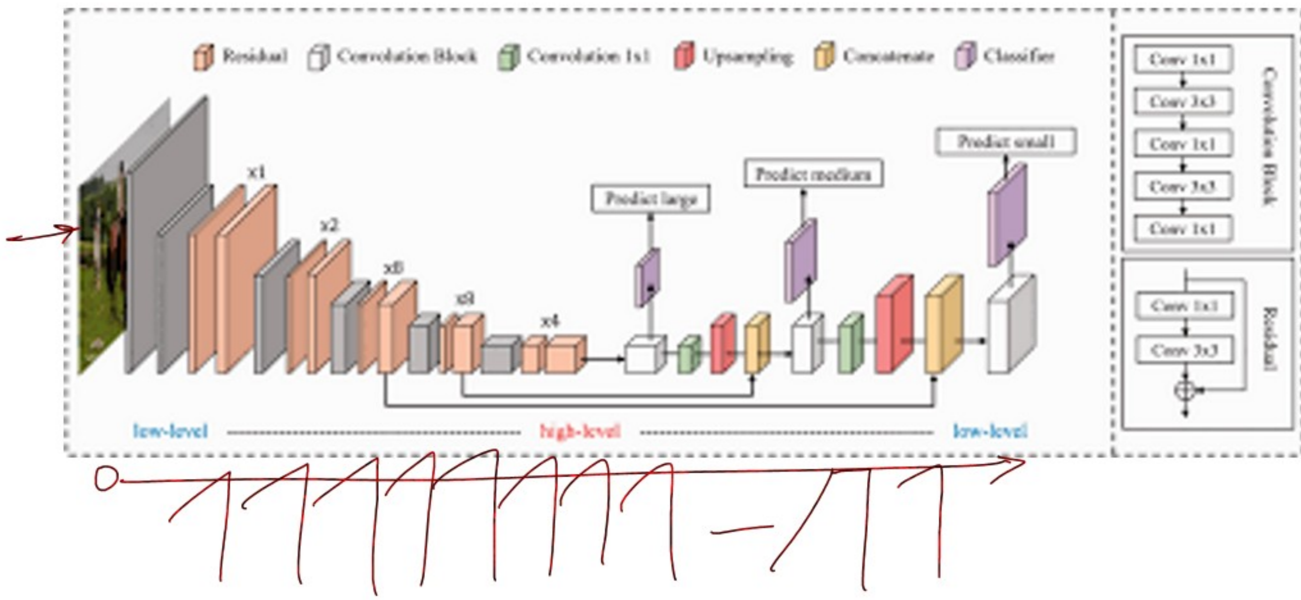
What is 'deep' in the deep learning?

- DL is a subset of ML (AI) that uses ANNs with multiple layers (deep networks) to model complex patterns in data.

[You only look once]
YOLO # Deep Convolutional Neural Network

YOLO Object Detection





Machine Learning Vs Deep Learning

*Pro-tip

comprehensive Comparison

Feature	Machine Learning (ML)	Deep Learning (DL)
Definition	Algorithms that learn patterns from data without explicit programming.	Subset of ML that uses deep neural networks to model complex patterns.
Data Requirements	Works well with small to medium datasets.	Requires large datasets for better performance.
Feature Engineering	Requires manual feature extraction by domain experts.	Automatically extracts features from raw data.
Model Complexity	Uses simpler models like Decision Trees, SVM, and Random Forests.	Uses complex neural networks (CNNs, RNNs, Transformers).
Training Time	Faster training (minutes to hours).	Slower training (hours to weeks).
Computational Power	Works on CPUs; does not require high-end GPUs.	Requires GPUs/TPUs due to heavy computations.
Interpretability	More explainable and interpretable.	Harder to interpret (black-box nature).

APC to store a

APC to Store a nice textbook

	Forests.	
Training Time	Faster training (minutes to hours).	Slower training (hours to weeks).
Computational Power	Works on CPUs; does not require high-end GPUs.	Requires GPUs/TPUs due to heavy computations.
Interpretability	More explainable and interpretable.	Harder to interpret (black-box nature).
Use Cases	Fraud detection, recommendation systems, structured data analysis.	Image recognition, speech processing, NLP, self-driving cars.
Generalization	Works well on structured/tabular data.	Works best with unstructured data (text, images, audio).
Cost	Lower cost due to lower hardware needs.	Higher cost due to expensive training and hardware.
Popular Libraries	Scikit-learn, XGBoost, LightGBM, H2O.ai.	TensorFlow, PyTorch, Keras, Theano.
Real-World Examples	Netflix recommendations, spam filtering, credit scoring.	Face recognition, self-driving cars, chatbots (ChatGPT).

DL → LLM (large Language Models)

→ unstructured data (text) ←

- ANN is a computational model inspired by the structure and function of the brain

[there are billions of neurons]

- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

Analysis

Here's the text from the image:

- ANN is a computational model inspired by the structure and function of the brain. (there are billions of neurons)
- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

Turing Test # Alan Turing → [Imitation Game]

GPT # 3.0 → Generative Pre-Trained Transformer

Transformer → is an advanced DL model - Self attention Mechanism (BERT)

570GB of text data ≅ 300 billions of tokens - will be discussed in NLP later

tokens → (is a unit of text - generally a word)

* Pro-tip *

Why is deep learning called so??

- DL is a subset of ML that uses ANNs with multiple layers (50-100 layers)

Input

a lot of

↓

Deep learning

↓

(to model complex patterns in data)

Deep learning Industrial Use-cases

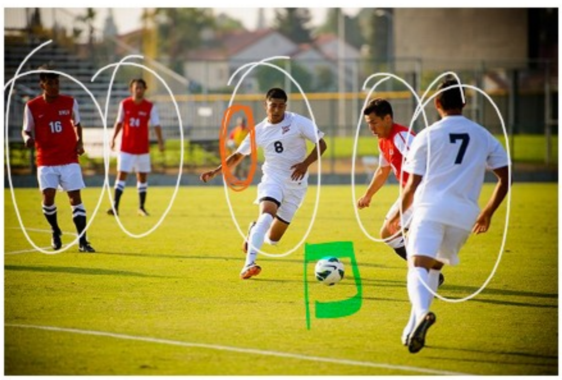
YOLO – You Only look Once # Objection detection algorithm

V3 → V12

de focussed image

↓

YOLO could identify.



HEALTHCARE

a) Medical Imaging AI models are being used in research to detect tumors, lesions or any other abnormalities in medical images. using CNN.

[MRI, CT, X-rays + patient's profile] → ??

What is the biggest reason for senior citizens to get blind?

[Diabetic Retinopathy]

↓

Diabetes leads to blindness

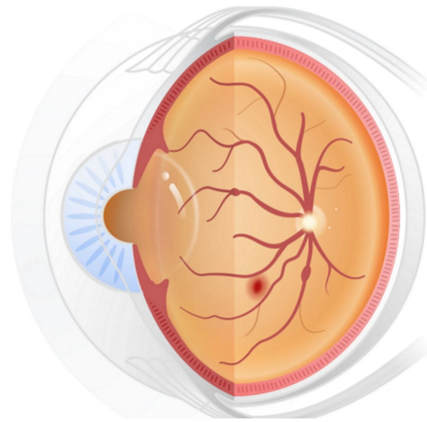
Diabetes leads to blindness

1.

Using AI to help doctors address eye disease

Helping doctors prevent blindness

Automated Retinal Disease Assessment (ARDA) is being used to help clinicians detect diabetic retinopathy, a leading cause of blindness, in India and across the world. With widespread adoption, perhaps millions of patients with diabetes could keep their vision in part to ARDA assisting doctors. Our research was published in JAMA and Ophthalmology. Additional research, published in Lancet Digital Health, showed that we can predict whether patients will develop diabetic retinopathy in the future, which can help doctors customize both treatment and eye screening frequencies for their patients. The solution is currently being evaluated in clinical studies in the United States as well as in Thailand. Learn more.



Developing a diabetic retinopathy screening solution

In research published in JAMA, Google's artificial intelligence accurately interpreted retinal scans to detect diabetic retinopathy. To do this, Google worked with a large team of ophthalmologists who helped us train the AI model by manually reviewing more than 100,000 de-identified retinal scans. This led to a development of an AI-based application called Automated Retinal Disease Assessment. This application can help doctors expand high-quality diabetic retinopathy screening programs in countries without enough eye specialists, such as India and Thailand.

The Indian EXPRESS
JOURNALISM OF COURAGE

Premium

Google AI tool for retinal scan can predict cardiovascular risk

The technology could reveal the heart's health condition after matching the eye scans with a matrix for cardiovascular risks. The algorithm has proved to be correct in 70 per cent of the cases where it has been tested so far.

Written by Anonna Dutt
Updated: July 2, 2023 12:31 IST

NewsGuard

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Why should Google tie-up with Aravind Eye hospital, Madurai?

Jagan N
Learn → Teach → Grow

October 15, 2023

In the recent Google I/O, there was an announcement about Google's tie-up with Aravind Eye hospital in Madurai to develop and deploy machine learning and artificial intelligence capabilities to help eliminate needless blindness.

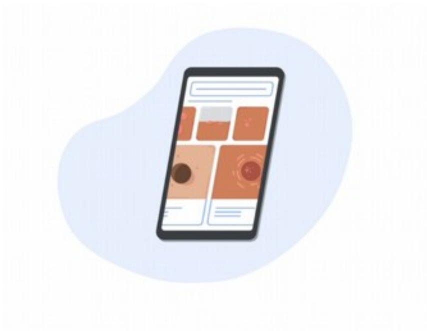
ARAVIND EYE CARE SYSTEM

We are collaborating with Aravind Eye Hospital in India, in our development and deployment of machine learning and artificial intelligence capabilities for retinal imaging, with a focus on diagnosing diabetic retinopathy, to help eliminate needless blindness.

(2.) Which continent is the most prone to skin diseases?

Australia (UV rays are more dangerous due to depleted ozone layer)

Africa



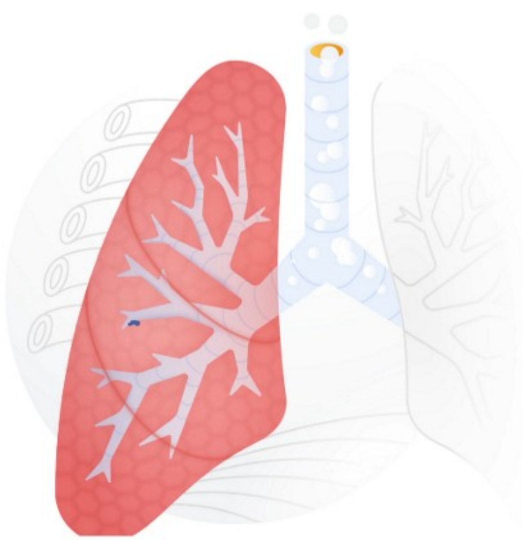
Improving access to skin disease information

Through computer vision AI and image search capabilities, we are developing a tool to help individuals better research & identify their skin, hair, and nail conditions. The tool supports hundreds of conditions, including more than 80% of the conditions seen in clinics and more than 90% of the most commonly searched conditions. The work was highlighted in both Nature Medicine and JAMA Network Open. Learn more

This product has been CE marked as a Class I medical device in the EU. It is not available in the United States. This research is not related to the DermAssist tool, which is no longer in development.

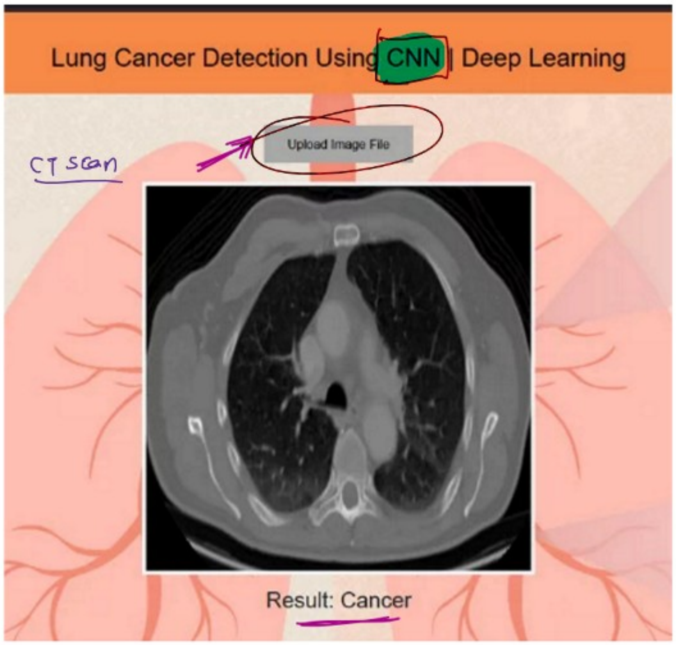
#3

Using AI to improve lung cancer detection



A promising step forward for predicting lung cancer

Lung cancer leads to over 1.8 million deaths per year world wide, accounting for almost one in five cancer deaths, and is the largest cause of cancer mortality. Our research, published in Nature Medicine, shows that deep learning may eventually help physicians more accurately screen for lung cancer and identify the disease even in incidental lung cancer detection workflows. Read the post

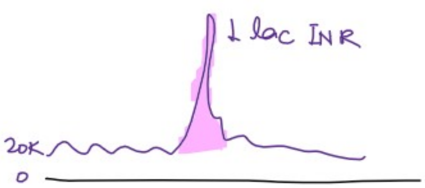


— APC to share the journal paper (pls remind me) ✓✓

FINANCE

Broadly speaking BFSI sector

Fraud Detection: To detect fraudulent transaction using autoencoders and also RNN / LSTM.



Reactive
↓
Preventive (Pro-active) → issue → (True Positive) (False Positive)
↓
Sophisticated [Predictive + Preventive + Prescriptive]

[No motivation provided for trading !!!]

Algorithmic Trading: To make trading decisions using some of the state of art algorithms → LSTM

STOCK??

Technical charts / Metrics / KPI → ML → DL (RNN → LSTM) → LLM
↓

Gen AI

Metrics | KPI → ML → DL (RNN → LSTM) → LLM

↓

(Agentic AI)

credit scoring | loan eligibility : To do the paperwork to access and assess credit score to understand the financial health and decide on the loan's eligibility.

- Payslips (last 3|6 months)
- bank statements
  - + 10 documents
- ITRs

+ Credit Score – CIBIL (XXX/900)

Analysis

Credit Scoring/Loan Eligibility: To do the paperwork to access and assess credit score to understand the financial health and decide on the loan eligibility - using RNN/LSTM + advanced NLP (LLM)

↳ handwritten documents

Law | court | Insurances | Govt or Public → there is ton of paperwork.

(TATA AIA)

Know your rights

↓

take informed decision

AUTOMOBILES

ADAS : Advanced Driver Assistance Systems

(1) Lane Keep Assist (LKA): it detects lane markings and warns or steers the car back into the lane.

(2) Park Assist

(real-time images) + (monitoring)

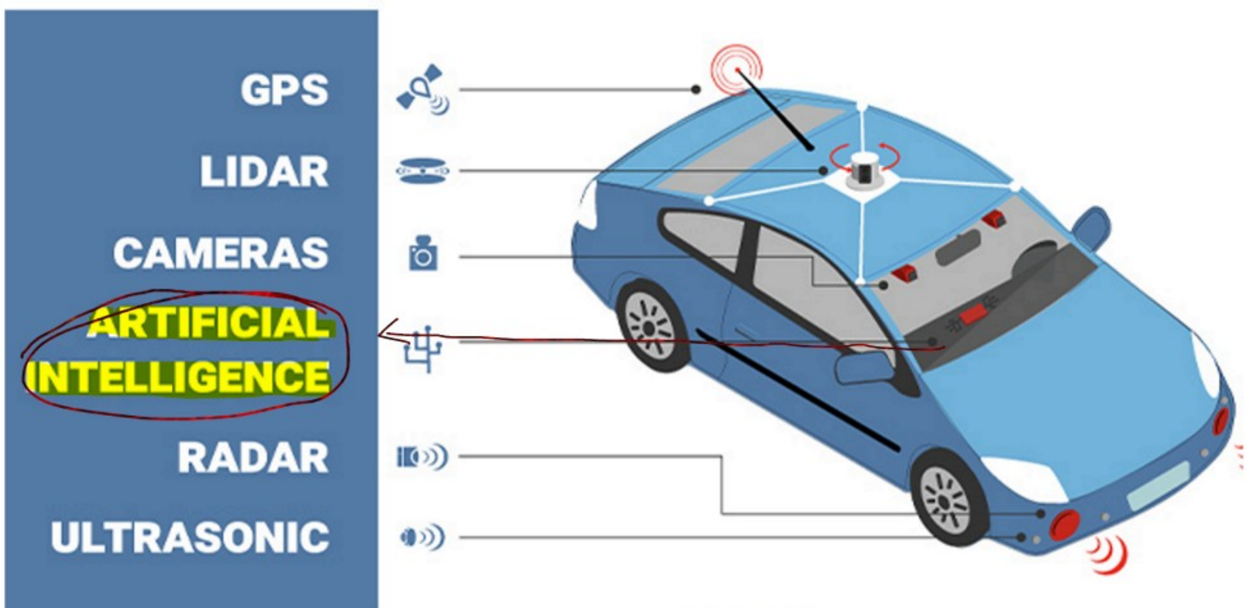
(2) Park Assist

(3) Driver Monitoring (fatigue detection)

+ (monitoring)

(self-driving cars)

- world class use-case of computer vision (CV)



Logistics

(drone-delivery)

(drone-delivery)

https://www.usatoday.com/story/tech/2013/12/01/amazon-bezos-drone-delivery/3799021/

Special Mentions

- Supply chain
- Quick E-commerce case-study
- Telecom : Preventive Asset Maintenance (PAM)

Airtel # Fraud | Scam detection & prevention
— Link will get deactivated

Data-centers → PAM

Module #① Introduction to deep learning (DL)

bonus

- Difference b/w ML, DL, AI, GenAI, LLM, Agentic AI, DS, RAG NLP → as a separate module.
- Use cases of DL in the industry
- (domain wise) ↔ examples